

Fraction Market with Bottle Caps

A low-cost multigrade activity for representing, comparing, and explaining fractions using familiar classroom objects.

GRADES 3 to 5	TIME 40 to 55 minutes	MATERIALS Bottle caps, paper, pencil	GROUPING Pairs or small groups
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Learning goals

By the end of the activity, learners should be able to:

- represent a fraction as equal groups or parts;
- compare two fractions using objects, drawings, or words;
- explain how they know which fraction is greater or whether two fractions are equivalent.

Teacher check	Ask learners to explain their model before accepting a numerical answer. Their explanation is the strongest evidence of understanding.
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Before class

Prepare 20 to 30 clean bottle caps per group. If bottle caps are unavailable, use seeds, pebbles, folded paper squares, or other safe objects that can be counted and moved.

Write these sample market orders on the board: one-half of 12 caps, one-third of 12 caps, three-fourths of 16 caps, and two-thirds of 18 caps.

ACTIVITY FLOW

Run the Fraction Market

STEP 1 Model	STEP 2 Compare	STEP 3 Explain	STEP 4 Create
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1. Warm-up: fair shares - 8 minutes

Give each pair 12 caps. Ask them to show one-half, one-third, and one-fourth of the set. After every model, ask: How many equal groups did you make? How many caps are in the named part?

2. Market orders - 15 minutes

Read one order at a time. Learners build the fraction, sketch what they built, and write the fraction. Let older learners create two correct models for the same fraction.

3. Compare two orders - 12 minutes

Pairs compare two models. They complete this sentence: I think ____ is greater than / less than / equal to ____ because _____. Encourage them to rearrange caps to prove the comparison.

4. Learner-created challenge - 10 minutes

Each pair creates one market order for another pair. The order must include a fraction, a total number of caps, and a way to check the answer.

Multigrade move

Grade 3: unit fractions. Grade 4: like and unlike denominators using models. Grade 5: equivalent fractions and explanation. Everyone uses the same materials, but the reasoning demand changes.

Show your fraction thinking

Name: _____ Grade: _____ Date: _____

Market order 1

Write the fraction, draw or describe your model, and explain how you checked it.

Market order 2

Compare two fractions. Use greater than, less than, or equal to, then explain why.

My challenge for another pair

Write a new fraction market order and the answer key.

Exit check

Show three-fourths of 12 using caps or a drawing. Write one sentence that explains why your model is correct.